Kailo Coaching Hub Audit

Product review & data-platform architecture

21 August 2026

Defining Product Goals

- **Team-native, coach-facing training platform.** The team is the unit; the coach is the primary user.
- **Stronger analytical tools and coach customization** than TrainingPeaks or FinalSurge — the analytics and the ability to tailor the surface are the differentiators, not feature parity.
- **A coach-determined path for athletes to interact** that *reduces* the need for athlete buy-in. The less work an athlete has to do, the better the product: participation should be near-passive by default, with the coach deciding how much the athlete is asked to do.

Next Steps (ordered)

1. **Restructure the hub into the five-tab layout** of §1.3 — Today, Upcoming, Team, Recruiting (v2), Settings — with the athlete page reached through the Team tab (roster, results, or the PR / progression boards) rather than as its own tab.
2. **Athlete objects with histories and attributes.** Each athlete carries a history (race and training paces over time) *and* a set of learned attributes — e.g. "performs better on hilly courses," "degrades more in rain / heat." These attributes are estimated from data, they change as more data arrives, and the set will grow. The data platform (§2) exists to populate and keep refreshing them.
3. **Text-to-structured-workout parsing.** The planner grid must turn "2 mile wu, 5 tempo, 3 cd" into a structured workout that can be analyzed post hoc; the current parser misreads such inputs (see the Planner and Workout planner audits).
4. **Weather upgrade.** Move the adjustment metric from dew point to wet-bulb globe temperature, and show conditions for four windows — this morning / afternoon, next morning / afternoon — to cover doubles.
5. **Data & enrichment platform build-out** (§2): divest from Athletic.net; per-race time, weather, and hill enrichment feeding the learned attributes.
6. **Recruiting** — a *v2 product.* Reserved in the tab structure but not built in v1.

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1 Product Audit

1.1 Two axes

The desktop hub is organized on two axes.

- **Vertical axis — thematic.** A left pane (top on narrow screens) of **five tier-0 objects**, each containing pages (Figure 1). The tier-0 objects are *Before Practice*, *Share with the Team*, *After Practice*, *The Roster*, and *Housekeeping*. Their labels are not clickable; the pages beneath them are. This axis is the spine of the audit below — every page gets an analysis, and every tier-0 object gets a general assessment.

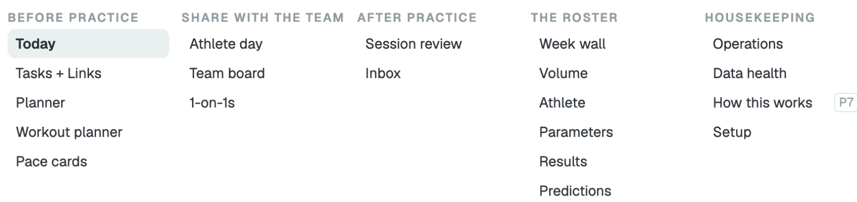

Figure 1: Vertical (thematic) axis as of August 21, 2026

- **Horizontal axis — temporal.** The coach's day, as a left-to-right flow. Each step is simply a **hyperlink** to one of the vertical-axis pages (Figure 2).

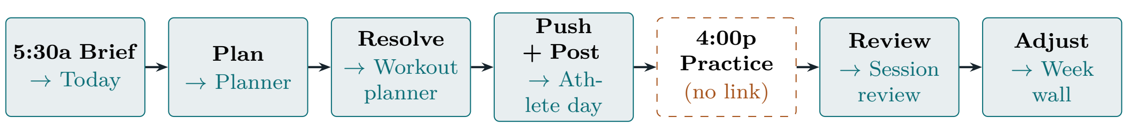

Figure 2: The horizontal (temporal) axis: the coach's day. Each step links to a vertical-axis page. **4:00p Practice currently links nowhere.**

The day cuts across the thematic tiers. The seven day-steps do not live inside one tier-0 object — they thread through four of them:

Day step	Links to	Tier-0 home
5:30a Brief	Today	Before Practice
Plan	Planner	Before Practice
Resolve	Workout planner	Before Practice
Push + Post	Athlete day	Share with the Team
4:00p Practice	— (nothing)	—
Review	Session review	After Practice
Adjust	Week wall	The Roster

[Future consideration should be made to decide whether information should be organized in a temporal or thematic axes.]

1.2 Vertical axis — page-by-page audit

Before Practice

Today [← day: 5:30a Brief]

This is the landing page (`kailo.fit/hub`) — a long stack of text widgets. Top widgets: *"[N] uploaded yesterday," "Available today," "Data flags," "Gone quiet," "Need a look,"* and *"Dew point at practice."* Below them: *"Where your attention goes"* (activities flagged abnormal — average heart rate high on an easy run, distance longer than expected), a *"Today's plan"* widget with a print option, and a practice conditions widget (*"Conditions this afternoon"*).

- **Keep high-quality, accessible print.** The plan's print option is good and should stay easy to reach throughout.
- **Conditions is buried and single-window.** It sits too far down, does not currently populate, and only shows the afternoon — athletes running *doubles* lose the morning window entirely.
- **No expand/collapse** on any widget, producing a very long page.

Assessment. Too much information, almost all of it text, with no visual hierarchy to separate what matters. Because this is the landing page, deep customization is essential.

Tasks + Links

This is a simple and considerably shorter page than the Today screen. It consists of two widgets: a basic task organizer and a place to store hyperlinks. The quick links widget prompts, "Add the ones you reach for weekly - entries, results, the conference site."

Assessment. A dedicated task page is likely useful, with a connected widget that can be added to a landing page. I don't see much long-term need for the "Quick links" section. Anything that would go there should be put in its proper place.

Planner [← day: Plan]

The planner page is a core aspect of the product. It has planning capabilities for the entire season (day, week, and entire training block). Weeks can be toggled one by one. There is a pin every Wednesday in this example. I am interested to know how the week grid works. Namely, does inputting text create a formatted workout? Ideally, one could write "Easy 45 minutes" or "2 mile wu, 5 tempo, 3 cd" in the grid and then recover an analysis of the workout based on how it was completed vs. predicted (easy pace was faster or higher-HR than normal; tempo pace is improving).

Assessment. This will be a core page going forward that should anchor a future "Upcoming" section. Technical considerations must be made to determine the degree to which text inputs can create structured workouts that can be analyzed post hoc.

Workout planner [← day: Resolve]

This seems to be a more zoomed-in version of the planner day above. It allows for creating workout pods, prescribing paces for a workout, and making weather/terrain adjustments. Again, there is some instructional language here that is just a bit over the top, such as "The question: one date, six lanes. I plug the workout in, the groups and paces come out, weather and terrain already applied." I'm not sure what the "What moved since last week" section corresponds to, unless it's simply product updates. There is a "Rules baked in" section, which is basically settings.

Assessment. Dew point is currently the listed weather metric used to determine weather adjustments. Based on the literature, we'd probably want this to move to wet-bulb globe temperature. The "Rules baked in" section should be removed and replaced with a link to the workout planner settings page, for organizational purposes. As above, we need to put in technical work to make the text-to-planner conversion work better. For instance, I tried to write "5 up, 2 tempo, 3 cd," and the current iteration put "2 tempo, 3 cd" as the recovery.

Pace cards

Basically the VDOT training paces. It is unclear what "Provisional" means here. It asks users whether they want "page" or "phone" views. The prescribed paces are: "This week," "Easy days," "Long run," "Weekly probe."

Assessment. This is again a core feature that should probably be linked in the training section but live under the Athlete pages. I don't understand the logic for why certain paces are excluded — namely recovery and tempo/LT.

Tier-0 assessment. In general this tier is too much of a catchall for the core elements of the product. This isn't necessarily a problem on its own. However, given how loaded this part is with features compared to the next two Tier-0 categories (Share with the Team and After Practice), it suggests there are better ways to organize these data and features to disperse them in a more coherent and balanced way. I would recommend creating a "Today" landing page and "Upcoming" planning sections that split up this data.

Share with the Team

Athlete day [← day: Push + Post]

This page allows you to see what a given athlete sees on their Kailo app for each day's practice.

Assessment. This page can likely be a submodule of existing planners as a "Preview" option.

Team board

This page is a grab bag of different planning and team-related tools. "This week" is the first tab and is similar to the Workout planner. PR Board, currently unpopulated, seems to be a way to assemble PRs from a given meet (or overall PRs — it is unclear which). The team document, also currently unpopulated, seems to be a place to record team rules/creeds/values. Race plans allow the user to add different goals/plans for specific athletes for specific races. You can also input other teams' results.

Assessment. This page has too many disparate elements that limit its usefulness. While there is a conceptual core, there are better places for race info, weekly planning, team rules, and team PRs. A specific team homepage would be useful, which could contain important division/conference information along with the team document. Race plans can definitely be improved by adding both predictive modeling for a given athlete (predicted pace range) and MCMC simulations to see how that athlete stacks up against expected competition.

1-on-1s

This is a central repository to prepare and document 1-on-1s. The structure of the meeting notes isn't great since there is no expand/collapse, so long notes would render the page extremely long.

Assessment. I think this is probably a service that only some coaches will use, given that many won't input notes on meetings, but it is a good feature to have. However, like other features in this section, I think it could be moved to a better location.

Tier-0 assessment. Generally, the features in this tier are all individually useful, but some read as redundant with other features, and all of them could be placed in more intuitive locations. I would recommend removing this as an organizing tier and splitting its data among future Today, Upcoming, and Team sections.

After Practice

Session review [← day: Review]

Currently this page is not populated at all, which makes it hard for me to fully assess. But it seems to be information about the most recent session.

Assessment. This page is, of course, extremely central to coaching. We can create a "Most recent session results" module for the "Today" screen with the information in this section. But we need a space to put historical session data (e.g., the session from five days ago).

Inbox

Some of these are repeated widgets from the current "Today" page (namely the "Needs a look" section, with some overlap in "Contact coverage"). The Reason ledger and Confounds sections are not fully populated for me, so I can't fully assess what's in them (though the latter seems to be some predictive or analytical system).

Assessment. Similar to the Share with the Team section, I think there is important information in this section that would be better served by being moved under a different header in a more optimal organizing strategy.

Tier-0 assessment. I echo what I said in the Share with the Team section: many of these features are useful but are either redundant or not optimally placed. I also think that coaches need a way to toggle certain features on or off, such as meeting tracking with students. If a coach doesn't manually input that information, the app may nag them that they're never meeting with students. As part of the customization layer, it would be useful to treat some of the functionality like a grand-strategy video game: add layers/features as you want, or simply use it as a more basic tool of roster visuals and workout planning.

The Roster

Week wall [← day: Adjust]

This seems to be another calendar overview of the training week. For me, it is populated by only two athletes.

Assessment. Important information, but redundant with other pages. Perhaps other pages on training programs could have various view options, so you could have a "week view" option like this.

Volume

A key page. Athlete name, 7-day and 28-day volume with wk/wk %, rest days, last upload, percent complete, directive, loop, and trend (in histogram form) all sit under the default "This window" pane. The second pane is "Plan & Goals," which has weekly volume goals and season peak-mileage goals (and down-week depth). There are unpopulated Actuals and Adherence panes.

Assessment. There needs to be a legend for certain terms or concepts such as Complete %, wk/wk %, directive, and loop. The trending inlay graph is pretty, but I'm not sure about its use. In general, lots of the information here is useful. But again, the Plan & Goals section would be better nestled in a planning section.

Athlete

This is basically a roster. It is a drop-down. Upon selecting an athlete, you get "The person file" with high-school info, dietary restrictions, and class schedules. There is also information about surveys and check-ins, notes, and the athlete's stated goals. The page continues to Paperwork, Training room, Recruiting file, Class schedule, and Medical and emergency contact.

Assessment. In general, this page is pretty long. This is more like an athlete's settings than a landing page. A more optimal landing page would be closer to a public roster: name, grade, gender, hometown, PRs, recent race results, and recent training (a week-over-week mileage graph, perhaps), along with some future-oriented things like goals. Dietary restrictions and class schedules could still be populated below, but some of the things currently on the page could simply be links to other sections instead of repeated widgets (such as 1-on-1 notes).

Parameters

This is a list of the parameters produced through modeling for each athlete. These include critical speed, Rx width, D-prime, anchors, speed reserve, and extrapolation.

Assessment. The listed critical speed (with the CIs attached) could be an optional widget added to athlete profiles. This page, generally, is probably best kept in the settings.

Results

As expected, a results page with tabs for "by meet," "season progression," "record board," "courses," "import," "goals," and "season review."

Assessment. As said above, some of the recent results and PRs could probably be included as (optional) widgets on the athlete profile page. But a central results repository is still important to include in the "Team" section. The import option should be a hyperlink to a pop-up, not its own tab. Some of the remaining information could likely be either reorganized or broken into different pages.

Predictions

This is a page devoted to assessing how accurate past race predictions were. Past results are shown ("Locked before the gun") in a widget, followed by a grading widget, a calibration widget, and a larger season replay widget.

Assessment. I'm not sure how central this page is. Perhaps because no predictions have been made yet — and I can't see current predictions — this page seems better nestled in the settings.

Tier-0 assessment. This is the section where not being able to click the tier-0 module name (The Roster) is most frustrating. I think the optimal flow would be a "Team" landing page with team name, picture, and conference/division information. This would then have a button for the roster, another for results, another for PRs, and perhaps season progression.

Housekeeping

Operations

This is a page for logistical/operational components of coaching, such as money or equipment information. It also allows .ics export of meet/training calendar information. The first two tabs — season board and this meet — seem to be repeats of existing data.

Assessment. This is probably better nestled under the Team label instead of the Housekeeping label, since I think the latter is better served as settings.

Data health

This is more of a classic settings page on data importation. The first widget (Coverage) shows runs carrying a heart-rate average, session uploads carrying lap marks, and athletes with a band prescribed today. This is followed by a widget on whose data is not arriving. There is then a long widget of where each athlete's data comes from (Strava vs. Garmin). There is also an extremely long section of flags from the data ("Things worth knowing about the data"). Finally, there is a widget showing who is missing paperwork.

Assessment. I'm not sure whether the paperwork described is medical clearances. If so, that information is too different from the data stuff here to be included. I think this could be recast as a "Data importation" settings menu.

How this works

An unpopulated introduction/explainer for the app.

Assessment. Obvious need.

Setup

Lots of settings panels here, which can all be broken into their own settings tabs. These include "Recruit-visit mode," which masks athlete names "so a campus visit sees the machinery and not the people." There are also settings about program information, team information (name, location, coaches/access settings), and data flags. There are also bulk import and export widgets.

Assessment. These need to be built out as their own settings menu. This includes recruit-visit mode, team settings (including team/practice location), coach settings, bulk import/export, and the various other settings listed above.

Tier-0 assessment. There is a clear need to make this tier into a more standardized settings menu that centralizes many of the options scattered throughout the current system.

1.3 Proposed structure: five main tabs

Collapse the sprawl into five tabs. This is a first pass, assembled from the per-page recommendations above.

1. **Today** — the landing page. Keeps what the current Today page gets right (the plan, with the print option easy to reach) and adds what the audit calls for: a "most recent session" module (from Session review), the needs-a-look flags, and practice conditions for *four* windows — this morning, this afternoon, next morning, next afternoon (fixing the doubles gap). Every widget collapsible, and the widget set coach-customizable.

2. **Upcoming** — planning, anchored by the Planner (day / week / training block) with the Workout planner as its zoomed-in day view. Absorbs race plans (from Team board) and the Plan & Goals material currently inside Volume.

3. **Team** — a team landing page (name, picture, conference/division information, the team document), with routes into:

   - **Roster** — the public-roster-style list; selecting an athlete opens their page (layout below);
   - **Results** — the central results repository (by meet, season progression, record board, courses), where every athlete name links to that athlete's page;
   - **PRs / season progression** — selecting an athlete goes to their page;
   - **Operations** — logistics: money, equipment, the .ics calendar export.

   The athlete page is deliberately *not* a top-level tab: the routes converge on it (Figure 5), so a coach reaches the same athlete object whether they come through the roster, the boards, or a result.

4. **Recruiting** — reserved as a tab, but a *v2 product*; not built in v1. (Recruit-visit mode — masking athlete names — ships in v1 under Settings.)
5. **Settings** — the standardized menu the Housekeeping audit calls for: data importation (Data health), team / coach / program settings, the workout planner rules, Parameters and Predictions, bulk import/export, recruit-visit mode, and the feature toggles (turn modules such as 1-on-1 tracking on or off).

Table 1 maps the proposal onto the current hub; the expanded plans below give the per-tab build detail.

Tab	Routes / contents	Absorbs from current hub
Today	glance row; conditions (four windows); today's plan + print; needs a look; most recent session; tasks	Today, Tasks + Links, Session review (as a module), Inbox (flags)
Upcoming	the planner at four zooms: block → week → day → athlete preview; race plans; plan & goals	Planner, Workout planner, Athlete day, Team board (race plans), Volume (goals), Week wall (as a view)
Team	landing page; Roster; Results; PRs / progression; Operations; the athlete pages	The Roster tier, Team board (team document, PR board), Operations, 1-on-1s (linked from athlete pages)
Recruiting	v2 placeholder	Recruiting file (from the person file)
Settings	data import; team / coach / program; workout rules; parameters; predictions; feature toggles; recruit-visit mode; bulk import/export; help	Data health, Setup, Parameters, Predictions, "Rules baked in," How this works

Table 1: The proposed five-tab structure, and where the current hub's pages land.

1.4 Expanded plans

First-pass build detail per tab. The athlete page gets the most, since it is the object the rest of the product hangs off.

Today

The landing page answers "what do I need to know before practice" in one screen (Figure 3):

- **Glance row** — uploaded yesterday, available today, data flags / gone quiet, needs a look (the renamed "Where your attention goes"), each a compact tile whose names link.
- **Conditions strip** — *four* windows (this morning, this afternoon, tomorrow morning, tomorrow afternoon) so doubles are covered, computed on wet-bulb globe temperature rather than dew point.
- **Today's plan** — with the print option kept one click away.
- **Most recent session** — the Session review material as a module, with historical sessions reachable from it.
- **Tasks** — the task organizer from Tasks + Links as a widget.

Every widget is collapsible; the widget set and its order are per-coach. Build order: the widget framework (collapse + persistence) first, then the conditions strip, then the recent-session module.

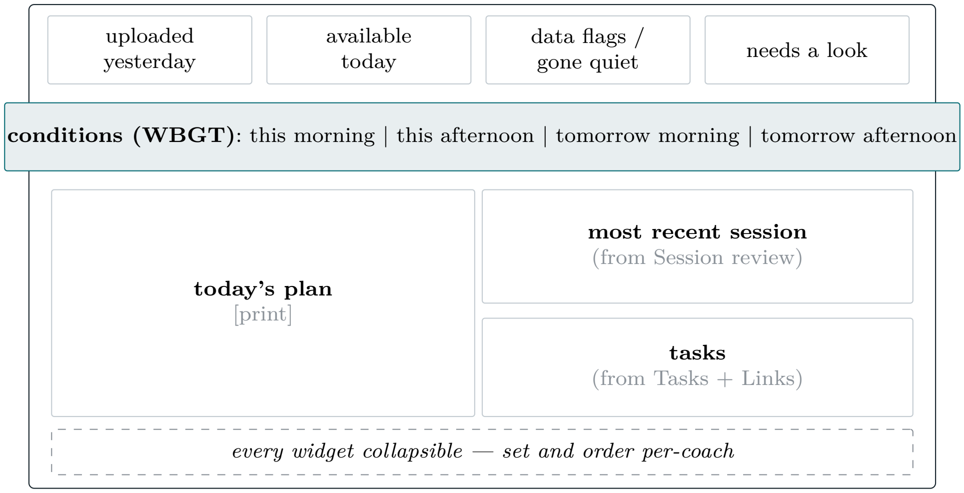

Figure 3: Today as one screen. The glance row links to people; conditions cover four windows so doubles are never blind.

Upcoming

One planning object at four zoom levels, rather than three separate pages (Figure 4): the season block (phases, goals, peak mileage, down-week depth — absorbing Volume's Plan & Goals), the week grid (Week wall becomes a view option here, not a page), the single day (the Workout planner's pods, paces, and weather/terrain adjustments), and the athlete preview (Athlete day as a "preview" button, not a destination).

- **Text-to-structured-workout parsing** is the load-bearing technical piece: "2 mile wu, 5 tempo, 3 cd" typed in the grid must become a structured workout that can be analyzed post hoc.
- **Race plans** move here from Team board and attach to meet dates: per-athlete goals plus predicted pace ranges, and eventually MCMC field simulations against expected competition.
- **Training paces** (the Pace cards content) surface here as links; the paces themselves live on the athlete page.

Build order: the four-zoom shell around the existing planner first; the text parser next; the race-plan predictive layer last.

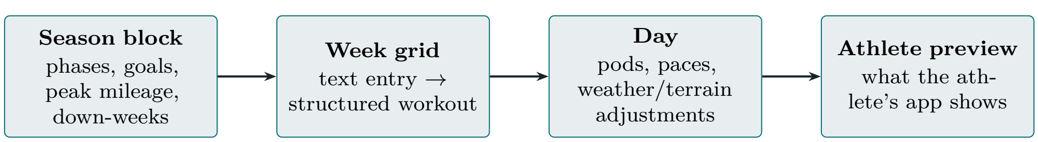

Figure 4: Upcoming: one planning object at four zoom levels. The current Planner / Workout planner / Athlete day split becomes zooming, not separate pages.

Team

A team landing page — name, picture, conference/division information, the team document (rules / creeds / values), and the PR board — with four routes (Figure 5): the Roster, the Results repository (import as a pop-up, not a tab), the PR / season-progression boards, and Operations (money, equipment, the .ics export of the meet and training calendar). Every surface that shows an athlete's name — roster row, result line, PR board entry — opens the same athlete page.

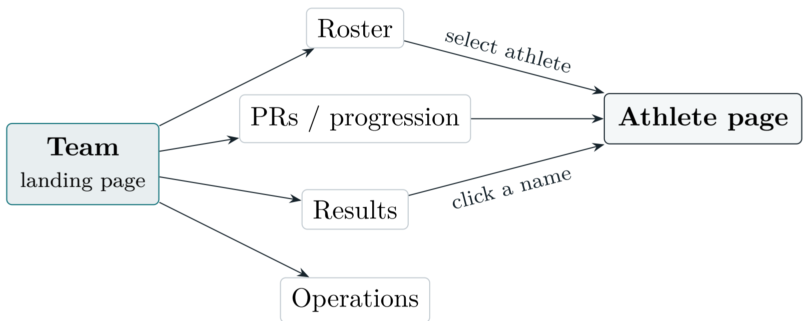

Figure 5: The Team tab. The athlete page is secondary: reached from the roster, the PR / progression boards, or any result line — all three converge on the same object. Operations stays team-level.

The athlete page (three tiers, top to bottom).

Identity (top, glanceable).
Name, gender, class, hometown, MileSplit and Athletic.net links, and PRs — the public-roster feel the audit of the current Athlete page calls for.

Secondary.
Recent race results, recent training (a week-over-week mileage graph), attendance / training log, past and current injuries, upcoming registered events (date, name, event), training paces, and stated goals. Person-file material (dietary restrictions, class schedules, paperwork) stays populated further down; anything duplicated elsewhere (e.g. 1-on-1 notes) becomes a link, not a repeated widget.

Advanced analytics (bottom).
The learned attribute models:

- **Hill Resilience** — is the runner better on flat or hilly courses? Built from course-elevation learning, MileSplit course rankings, FloRatings A/B/C/D, TullyRunners speed ratings, joshm.dev's Crystal Springs Pace Factor, and the golf-slope-rating analog.
- **Weather Impact** — how much adverse weather degrades this runner. The constraint: exact per-race time is needed, and getting it at scale from Athletic.net is not viable (see §2).
- Additional analytical models as desired.

Build order: the identity and secondary tiers ship first — they display data already held. The analytics tier lands as the platform models mature (§2), each model appearing as an optional widget rather than a blank promise.

Recruiting (v2)

Reserved as a tab but not built in v1; recruit-visit mode (masking athlete names) ships in v1 under Settings. When v2 comes, the tab inherits the recruiting file from the current person file, and the HS ↔ college linkage of §2 is its natural engine: a coach looks up a prospect's verified HS results and course-adjusted marks rather than a self-reported PR.

Settings

One standardized menu (Figure 6) replaces the Housekeeping tier: data importation from Data health, the Setup panels split into their own tabs, the Workout planner's "Rules baked in," the Parameters and Predictions pages, feature toggles (the grand-strategy layering — run the product as anything from roster-plus-planner up to the full analytics surface), recruit-visit mode, and bulk import/export, with "How this works" as the help entry. If the paperwork tracked in Data health is medical clearances, it moves to the Team tab's roster as a who-is-missing-paperwork widget — it is compliance, not data flow. This tab is a mechanical regrouping with no new functionality, which makes it a good early milestone.

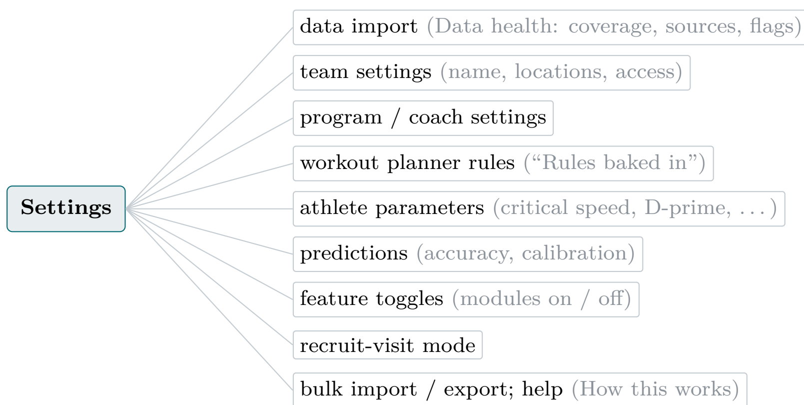

Figure 6: The Settings menu, centralizing what is currently scattered across Housekeeping, the Workout planner, and the athlete sub-pages.

1.5 Wording

Much of the copy reads as too cute, too artsy, or too AI-generated, and should be plainer. Examples:

Current	Plainer
"One line from an athlete at 3:47 is worth more than finding out at 4:10 in the parking lot."	Cut, or state the actual point.
"Where your attention goes" — "ranked, and every name is a door."	"Needs a look," with names that link.
"Yesterday, rolled up"	"Yesterday, summarized."

2 Data and Enrichment Platform

The athlete attributes in §1 (hill resilience, weather impact, and the models to come) are only as good as the data feeding them. This section lays out how that data is sourced, enriched, and linked — and why the platform must **divest fully from Athletic.net** as a scale dependency.

2.1 The Athletic.net problem

Athletic.net has the broadest US high-school coverage, and the project was built on it. But it is not a trustworthy dependency at scale:

- **It blocks us.** Cloudflare bot management returns `cf-mitigated=challenge` on sustained crawling. Measured over weeks: $\leq$ 1–2 req/s ran clean for days; 3 req/s drew 24-hour blocks on two consecutive days; volume near 30,000 requests/day drew challenges regardless of rate. Exit-node rotation across six IPs did not prevent it — the block is on the traffic *pattern*, not the address.
- **No sanctioned access.** No API, no developer program, no data-partnership channel.
- **The data itself has defects** we must work around — relay results collapse to one athlete per squad; per-race timestamps are all `T00:00:00`, so it cannot supply the race times the Weather Impact model needs anyway.

Assessment. Athletic.net stays as a redundancy check and for the states already captured, but every *new* pipeline is designed against sources that answer reliably — TFRRS, MileSplit, and static timing archives. The three diagrams below are that design.

2.2 (1) Roster, meet, and team information

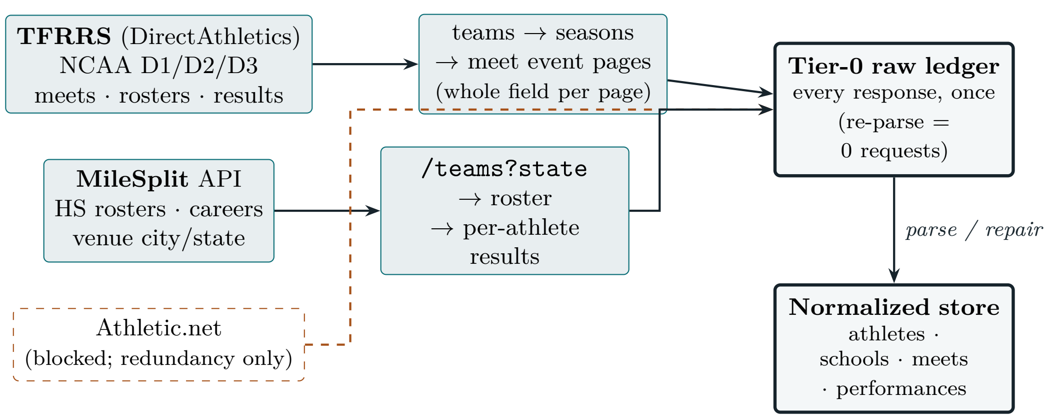

Figure 7: Roster/meet/team sourcing. Every response tees to a raw ledger before parsing, so a parser fix replays at zero request cost. Athletic.net (dashed) is demoted to redundancy.

How it works, per source.

- **TFRRS (college).** Seeded from the seven NCAA-division region pages, walk teams → seasons → meets. A single meet *event* page carries the whole field (place, athlete, class year, team, mark), so one fetch covers every school in a race. Corpus: 8.1M rows, all three divisions, XC + indoor + outdoor, 2021–2026.
- **MileSplit (HS).** `/teams?state=xx` then the proven per-team, per-athlete ingest path, smallest state first. Polite by default (rate-limited, ledger tee). Note the enumeration cap: `/teams?state` returns at most 999 rows, so the largest states need a per-county or meet-harvest route for full coverage.
- **Divest path.** New states come from MileSplit; existing Athletic.net captures (CA/TX/FL/NY) are retained and merged, but no new crawl depends on that host.

2.3 (2) Course weather and hill information

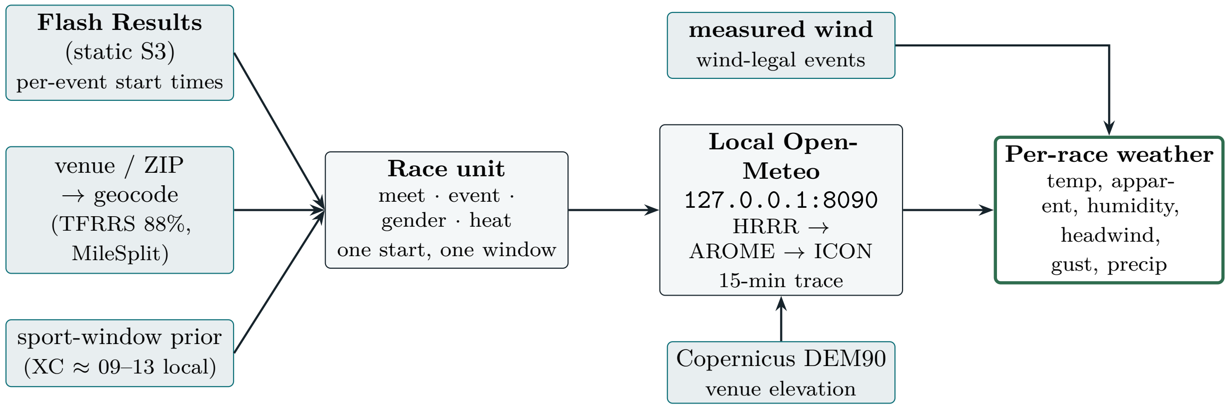

Figure 8: Weather + elevation per *race*, not per meet. The 11 am women's JV and the 3 pm men's varsity get different windows. Wind for sprints comes measured from the results themselves.

Hill / course difficulty (three complementary signals):

1. **Empirical ALS course offsets (built).** A two-way log-time model — log(equiv5k) = ability+ offset, i.e. an athlete-season term plus a course term — solved by alternating least squares over athletes who race multiple courses. Validated split-half $r = 0.98$; recovered Crystal Springs (+72s) and Woodbridge (−50s) with no external input. It measures what terrain *costs*, but bundles hills with footing and weather.
2. **External course ratings (cross-check + decomposition).** TullyRunners speed ratings, MileSplit course rankings, FloRatings A/B/C/D, joshm.dev Crystal Springs Pace Factor, golf-slope analog. Joining these and correlating with the ALS offset validates our number and separates terrain-driven from conditions-driven difficulty.
3. **True elevation gain (targeted).** GPS/GPX course traces through the DEM give cumulative climb per km — the only physical hill metric, but per-course external data. Reserve for marquee courses.

Assessment. Once weather is attached per race, re-fit the ALS *net of the weather term*: the course offset then stops absorbing that day's heat and becomes a terrain-only score — the cleanest hill metric derivable without a GPS trace.

2.4 (3) Linking college and high-school runners

There is no crosswalk: zero athletes carry both an Athletic.net id and a TFRRS id, and MileSplit's `tfrrsId` field is null on every bio examined. Linkage is therefore *probabilistic*, and the honest deliverable is the ceiling.

The route to a real crosswalk. TFRRS athlete pages carry a literal *"High School:"* field on 1.5% of archived pages (roster pages do not). Crawling athlete pages directly — rather than inferring identity from names — upgrades the linkage from probabilistic to keyed for the athletes who have it, and is the recommended next step for the recruiting and progression models.

2.5 Build order

Steps that cost nothing against any host come first:

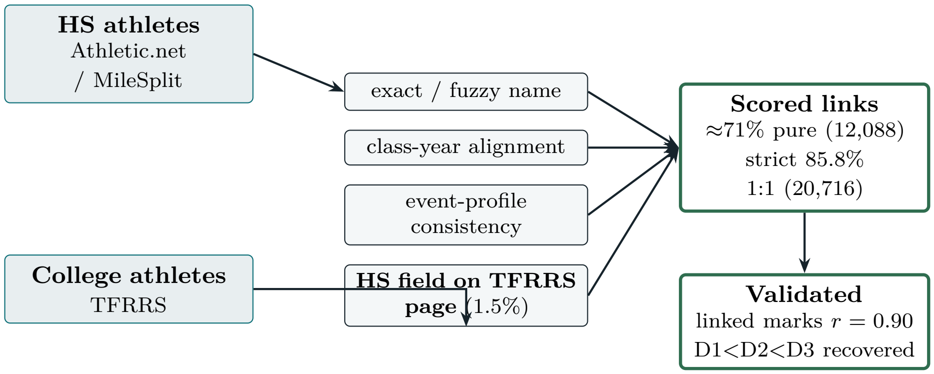

Figure 9: Probabilistic HS→college linkage. No exact key exists; a scored combination of name, year and event profile reaches ∼71% precision, and TFRRS athlete pages carry a literal High School field (1.5% of pages) that opens a path to a *real* crosswalk by crawling those pages directly.

1. Venue + geo columns; reparse venue from the archived TFRRS/MileSplit payloads. **Zero requests.**
2. Wind column on performances; backfill from archived Athletic.net payloads (wind-legal events). **Zero requests.**
3. Flash Results time crawler — new host, static S3, its own polite profile and ledger. Start with the NCAA championship/regional set and measure the match rate before scaling.
4. Weather adapter: emit a per-race manifest, run the existing `~/weather` Open-Meteo pipeline unchanged, reduce to race-window aggregates.
5. Course difficulty: ship the ALS offset, join external ratings as a check, then re-fit net of weather for a terrain-only score.